

Krupavathi Katakam

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Professional Summary

Software Engineer with experience in full-stack development, backend systems, AI/ML integration, and high-performance computing. Built LLM-powered applications and optimized performance for real-world workloads. Proficient in designing reliable architectures, integrating APIs, and developing data-driven solutions.

Experience

High Performance Computing Intern

April 2025 – June 2025

Centre for Development of Advanced Computing (C-DAC), Pune

- Achieved **3× speedup** in telecom data processing by optimizing **parallel execution** using **OpenMP** and **Intel MKL** for large-scale workloads.
- Reduced computation time by **60%** using **GPU-accelerated pipelines** for large-scale telecom dataset analysis.
- Developed **network traffic visualization system** and improved throughput via **data preprocessing** and **load balancing** on **HPC** systems.

Projects

Flowvy - AI Powered Workflow Builder

[Live](#) — [Link](#)

Tech Stack: Next.js, TypeScript, React Flow, PostgreSQL, Prisma, Gemini API

- Architected an **LLM-powered DAG workflow engine** with **type-safe execution** and node-based orchestration
- Accelerated workflow execution by **40–50%** through **parallel scheduling** using Trigger.dev for multimodal pipelines
- Designed **scalable backend systems and REST APIs** for workflow orchestration and execution tracking
- Implemented **secure authentication, persistence, and execution tracking** using Clerk, PostgreSQL, Prisma, and Zustand, enabling **workflow-run persistence, node-level logging, undo/redo functionality**.

Text-to-SQL - LLM-Powered Query Engine

[Link](#)

Tech Stack: Python, SQL, LLM APIs, RESTful APIs

- Built an **LLM-based system** to translate natural language into SQL, supporting **multiple query types** across relational databases
- Designed **schema-aware prompting and validation pipelines**, improving query correctness.
- Implemented backend modules for **query execution, error handling, and structured result formatting**.

MediZone – Integrated Healthcare Service Platform

[Link](#)

Tech Stack: Flutter, SQLite

- Developed a **cross-platform mobile healthcare app** with **4+ core modules** including doctor discovery, appointment booking, medicine ordering, and lab services
- Designed a **centralized order management system** to track **appointments, reports, and transactions**
- Implemented an **offline-first architecture** using SQLite, ensuring **reliable access in low-connectivity environments** and seamless data persistence

Education

Vasireddy Venkatadri International Technological University, Guntur

2023 – 2027

B.Tech in Computer Science and Artificial Intelligence & Machine Learning

9.1/10 CGPA

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Data Analytics & Visualization, Computer Networks, Object-Oriented Programming, Machine Learning, Data Science.

Skills

- Programming Languages:** Java, Python, JavaScript, C, C++
- Web & Backend Development:** Next.js, React, HTML5, CSS, RESTful APIs
- Databases:** PostgreSQL, SQLite, MySQL
- Tools:** Git, GitHub, VS Code
- Soft Skills:** Problem Solving, Team Collaboration, Communication, Time Management

Achievements

- Ranked in the **top 0.4%** in **Naukri Campus Young Turks 2025**, showcasing strong problem-solving skills [Link](#)
- Selected as a **National Semi-Finalist** in **Flipkart GRiD 7.0**, solving real-world coding challenges [Link](#)
- Ranked in the **top 3%** in **AINCAT 2025**, demonstrating strong aptitude and reasoning ability [Link](#)
- Achieved **Rank 92** on **GeeksforGeeks** by solving **1000+ DSA problems** [Link](#)